POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Operating Systems (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain operating systems as a resource manager. Describe batch, time-sharing and distributed operating systems in brief. 8

   b) Define Starvation in context to priority CPU scheduling algorithm. How would a Round Robin Scheduling algorithm behave when its time quantum is increased significantly? Explain with an example. 7

   OR

   What is the race condition and critical section problem? Define semaphore and use it to solve producer consumer problems.

2. a) Suppose that the following processes with their burst time arrive for execution at the times indicated. Solve the following 8

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	1

   i. Draw Gant chart for FCFS, SJF(non-preemptive) and SRTN
   ii. Find AWT and ATAT.
   iii. Determine which one among them would be the worst algorithm.

   b) What are classical IPC problems? Explain reader writer problem. 7

3. a) Differentiate between internal fragmentation and external fragmentation? Why do we need virtual memory? Explain the translation of logical address to physical address using paging with necessary diagrams. 8

   b) What is page table? Explain hierarchical and hased page table. 7

   OR

   Explain thrashing. What is the purpose of a translation lookaside buffer(TLB)? In a fixed partitioning scheme, what are the advantages of using unequal-size partitions?

4. a) Consider the following page reference strings: 2, 3, 4, 5, 3, 2, 6, 1, 3, 2. How many page faults would occur for each of the following page replacement algorithms assuming 3 pages a frame? In each case calculate fault ratio. 8
   i. FIFO
   ii. LRU
   iii. Second chance

   b) Differentiate between interrupt I/O and Programmed I/O. Describe a DMA controller with necessary diagrams. 7

5. a) Consider following snapshot of a system 7

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	1	1	2	1	1	2	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	5	0	6	7			
P4	0	0	2	0	6	6			

   Answer the following questions using Banker's algorithm:
   i. What is the content of the matrix needed?
   ii. Is the system in a safe state? Also find the safe sequence.

   b) Describe the file descriptor. Explain the linked list file allocation method. 8

6. a) Differentiate between process and threads. What are different multithreading models? Explain. 7

   b) What is cloud operating system? Discuss its characteristics and advantages. 8

7. Write short notes on: **(Any two)** 2×5
   a) The Shell and Virtual Machine
   b) Window vs Linux
   c) Monolithic kernel and micro kernel

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